

Lab 5: Regularization

We will use the `Hitters` data set in the `ISLR` package to predict `Salary` for baseball players.

```
library(ISLR)
library(tidyverse)
library(knitr)

str(Hitters)
```

```
## 'data.frame':   322 obs. of  20 variables:
## $ AtBat      : int   293 315 479 496 321 594 185 298 323 401 ...
## $ Hits       : int    66 81 130 141 87 169 37 73 81 92 ...
## $ HmRun      : int     1 7 18 20 10 4 1 0 6 17 ...
## $ Runs       : int    30 24 66 65 39 74 23 24 26 49 ...
## $ RBI        : int    29 38 72 78 42 51 8 24 32 66 ...
## $ Walks      : int    14 39 76 37 30 35 21 7 8 65 ...
## $ Years      : int     1 14 3 11 2 11 2 3 2 13 ...
## $ CAtBat     : int   293 3449 1624 5628 396 4408 214 509 341 5206
## ...
## $ CHits      : int    66 835 457 1575 101 1133 42 108 86 1332 ...
## $ CHmRun     : int     1 69 63 225 12 19 1 0 6 253 ...
## $ CRuns      : int    30 321 224 828 48 501 30 41 32 784 ...
## $ CRBI       : int    29 414 266 838 46 336 9 37 34 890 ...
## $ CWalks     : int    14 375 263 354 33 194 24 12 8 866 ...
## $ League     : Factor w/ 2 levels "A","N": 1 2 1 2 2 1 2 1 2 1 ...
## $ Division   : Factor w/ 2 levels "E","W": 1 2 2 1 1 2 1 2 2 1 ...
## $ PutOuts    : int   446 632 880 200 805 282 76 121 143 0 ...
## $ Assists    : int    33 43 82 11 40 421 127 283 290 0 ...
## $ Errors     : int    20 10 14 3 4 25 7 9 19 0 ...
## $ Salary     : num   NA 475 480 500 91.5 750 70 100 75 1100 ...
## $ NewLeague  : Factor w/ 2 levels "A","N": 1 2 1 2 2 1 1 1 2 1 ...
```

1 Data Processing

1. Remove records with missing data. Create a new (complete) version of your data set. (Hint: `drop_na` in `tidyr` could be helpful.)

2. You may need to create dummy variables for categorical variables in your recipes.
`step_dummy(all_nominal_predictors())` is a good way to do this.
3. You may need to standardize all variables in your recipes.
`step_normalize(all_predictors())` is a good way to do this.

2 Ridge Regression

The `linear_reg()` specification can perform both ridge regression and the lasso. This is done with the specification of a parameter `mixture`. If `mixture = 0` then a ridge regression model is fit and if `mixture = 1` then the lasso is fit. Here is an example for ridge regression with $\text{penalty} = \lambda$:

```
ridge_spec <- linear_reg(mixture = 0, penalty = lambda) |>
  set_mode("regression") |>
  set_engine("glmnet")
```

1. Create a vector of λ values from $\lambda = .01$ to $\lambda = 10^10$ of length 100.
2. Fit a ridge regression model for each λ in your grid. Be sure to normalize your predictors.
3. Make a line plot of coefficient corresponding to each λ . You should have an individual line for each variable with coefficient value on the y -axis and λ on the x axis. What happens to your coefficients as λ increases?
4. Perform 10-fold cross validation and get an estimate of the test MSE for each λ in your grid. Which λ would you choose and why? (Hint: look at the `tune` package for a fast way to do this.)

3 Lasso

1. Fit the lasso model for each λ in your grid.
2. Make a line plot of coefficient corresponding to each λ . You should have an individual line for each variable with coefficient value on the y -axis and λ on the x axis. (Hint: `coef` may be a useful function). What happens to your coefficients as λ increases?
3. Perform 10-fold cross validation and get an estimate of the test MSE for each λ in your grid. Which λ would you choose and why?